### 3.1.1 Data in

Elasticsearch needs data before it can provide answers to queries. Data can be indexed into Elasticsearch from multiple sources and in various ways: extracted from a database, copied from file systems, loaded from other systems (including real-time streaming systems), and so on.

Figure 3.1 shows data being ingested into Elasticsearch via three data sources:

- *Database*—Applications usually store data in a database as an authoritative system of record. We can prime Elasticsearch with data fetched from a database on a batch or near-real-time basis. As the data shape in a database may not be exactly what Elasticsearch expects, we can use an extract, transform, load (ETL) tool (such as Logstash—a data processing tool from the Elastic suite) to transform and enrich the data before it is indexed into Elasticsearch.
- *File store*—Applications, virtual machines (VMs), databases, and other systems spit out tons of logs and metrics data. This data is important for analyzing errors, debugging applications, tracing user requests, and auditing. It can be held in physical hardware storage or cloud locations like AWS S3 or Azure files/blobs. This data must be brought into Elasticsearch for search, analytics, and storage. We can use tools such as Filebeat (a data shipper with the sole aim of getting data from files into destinations like Elasticsearch) or Logstash to dump data into Elasticsearch for search, debug, and analysis purposes.

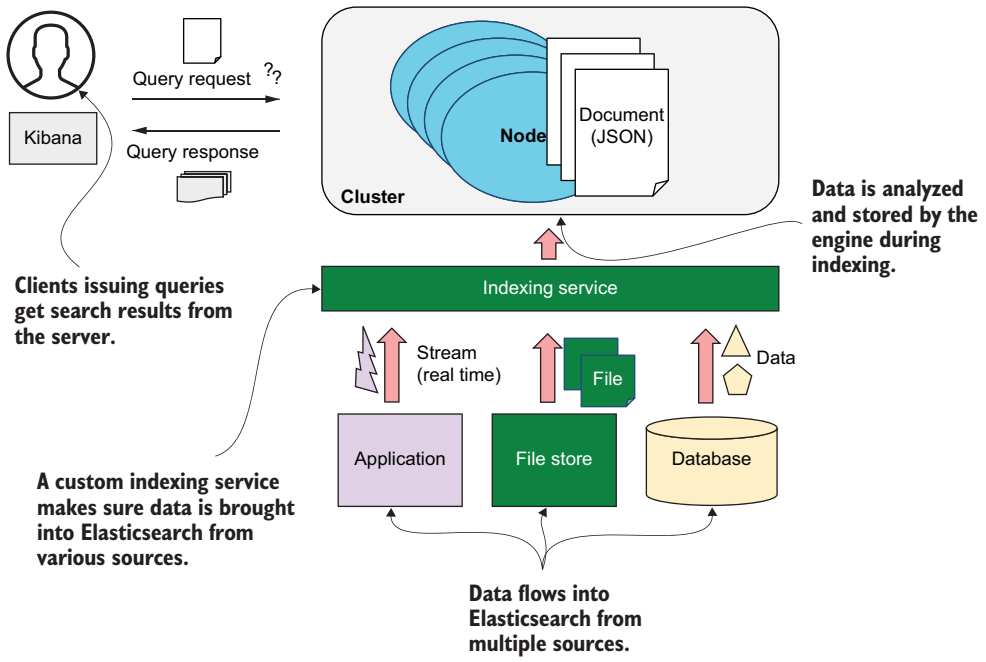



Figure 3.1 Priming Elasticsearch with data